Assignment No: 6

Jenkins Pipeline with Automated Testing for a Web Application

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Aim

To configure a Jenkins Pipeline with automated tests so that the pipeline fails automatically when any test case fails.

Output

1. Run tests locally

```
ronak@ronak:~/devops-task-manager$ npm test
> task-manager@1.0.0 test
> node test.js
Starting automated tests...
TEST PASSED: app.js exists
TEST PASSED: package.json exists
TEST PASSED: public/index.html exists
All automated tests passed.
```

2. Push Jenkinsfile and tests

```
ronak@ronak:~/devops-task-manager$ git add .
ronak@ronak:~/devops-task-manager$ git commit -m "Add Jenkins pipeline and automated tests"
[main 9e1f4c6] Add Jenkins pipeline and automated tests
 2 files changed
ronak@ronak:~/devops-task-manager$ git push origin main
```

3. Jenkins job

```
Job Name      : task-manager-pipeline
Type          : Pipeline
Definition    : Pipeline script from SCM
SCM           : Git
Branch        : */main
Script Path   : Jenkinsfile
```

4. Successful pipeline (Console Output)

```
[Pipeline] stage (Checkout)
[Pipeline] stage (Install Dependencies)
+ npm install
[Pipeline] stage (Build)
+ npm run build
[Pipeline] stage (Automated Testing)
+ npm test
Starting automated tests...
TEST PASSED: app.js exists
TEST PASSED: package.json exists
TEST PASSED: public/index.html exists
All automated tests passed.
Finished: SUCCESS
```

5. Failed pipeline (public/index.html renamed)

```
ronak@ronak:~/devops-task-manager$ git commit -m "Test pipeline failure"
ronak@ronak:~/devops-task-manager$ git push origin main
[Pipeline] stage (Automated Testing)
+ npm test
Starting automated tests...
TEST PASSED: app.js exists
TEST PASSED: package.json exists
TEST FAILED: public/index.html not found
script returned exit code 1
Finished: FAILURE
```

6. After restoring the file

```
[Pipeline] stage (Automated Testing)
All automated tests passed.
Finished: SUCCESS
```

Stage view

```
Checkout | Install Dependencies | Build | Automated Testing
  PASS   |        PASS          |  PASS |   PASS   (Build #3)
  PASS   |        PASS          |  PASS |   FAIL   (Build #2)
  PASS   |        PASS          |  PASS |   PASS   (Build #1)
```

Conclusion

Automated tests were integrated into a Jenkins pipeline; a failing test automatically failed the build, ensuring code quality.